Lecture 4

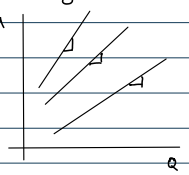
Thursday, 19 January 2023 1:32 PM

Open circuit potential, V_{oc}, potential diff b/w electrodes when $R_{load} \gg R_{cell}$, $I=0$

Faraday's law of electrolysis

1) Mass of product formed/consumed in e-chem rxn is directly proportional to charge passed

$\frac{m}{Q} = z$



linear irregardless of material

2) For a given amt of charge passed on electrode of different material, amt of product formed/consumed is proportional to electrochem equiv. weight of that product; $E = \frac{M}{n \cdot N_A}$

$m = x\,E = \frac{x}{n N_A}\frac{M \cdot e}{}$; e is fundamental charge

↓ Amount of e^- passed

ex is amount of charge passed Q

eN_A is Faraday's constant

$F = 96485.3 \frac{C}{mol\,e^-}$

$\therefore \boxed{m = \frac{QM}{nF}}$ → If we know amount of charge passed, we know how much mass is formed

$\frac{-\beta F V}{RT}$

$\frac{C}{mol \cdot e} \cdot \frac{K\,mol}{J} \cdot \frac{1}{K} \cdot V \Rightarrow \frac{CV}{J}$

Rates:

$I = \frac{\text{Charge}}{\text{time}}$ [=] $\frac{C}{s}$ [=] A

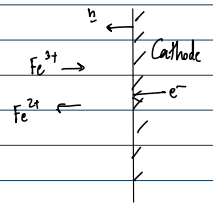
Faraday's law as a rate: $\frac{\frac{m}{M}}{s} = \frac{Q/s}{nF} \Rightarrow \boxed{r = \frac{I}{nF}}$ [=] $\frac{mol}{time}$

↓ mol of e^- / mol of product

→ 96485 $\frac{C}{mol \cdot e}$

Flux @ electrode surface

↓ vector quant.



$Fe^{3+} + e^- \rightleftharpoons Fe^{2+}$

↳ Flux balance: $Flux_{Fe^{3+}}$ ($\bar{N}_3$) = $Flux_{e^-}$ ($\bar{N}_e$) @ SS

↳ Molar Flux: $\left[\frac{mol}{cm^2 \cdot s}\right] = N_j$

$\vec{n} \cdot \bar{N}_3 = -\bar{n} \cdot \bar{N}_e = -\bar{n} \cdot \bar{N}_3$; $i = \frac{I}{Area}$ ⇒ Current Density

i/F ⇒ Cathodic current < 0

For a generic rxn, w/ stoich coeffs s_j:

$\sum s_j N_j \rightleftharpoons n \cdot e$

↓ (+ for products)

$\boxed{\bar{n} \cdot \bar{N}_j = -\frac{s_j}{nF} i}$ [=] $\frac{mol}{cm^2 \cdot s} = r' = \frac{-s_j\, i}{nF}$

↳ Faraday's Law as flux

} The minus sign for i would then mean it is Cathodic Current density.

→ Faraday's Law as flux density.

Electrode Potentials

$\Delta G = -nFV$; potential ~ ΔG } of an entire cell

V ↓ reversible cell potential

→ We care about potential of one electrode (usually working electrode)

↳ Counter electrode must exist, but it may complicate our potential measurements (since if we don't know the reactions happening at our counter, then we don't know if we are referencing to what)

↳ Therefore, we have to reference our WE potential to a 3rd electrode w/ a stable + defined potential → RE ↓ does not participate in the cell reactions

ex: $Ag_{(s)} + Cl^-_{(aq)} \rightarrow AgCl_{(s)} + e^-$

Electrochemical Potential

~ similar concept to chemical potential ; $\mu = \left.\frac{\partial G}{\partial n_j}\right|_{T,P,n_{i \neq j}}$

Charged species involved interact w/ electric field → Electrochemical potential

Fick's Law (Mass Transport)

Flux = convection + diffusion + electrical migration (for charged species)

↳ In dilute soln.: diffusion is very imp & defined by Fick's Law

$$\bar{N}_j = -D_j \cdot \nabla c_j \quad ; \quad D \approx 10^{-5}\ cm^2/s \text{ (in aq. systems)}$$

D = diffusivity

Ohm's Law (Current transport)

In well-mixed electrolyte, ohm can describe current flowing thru electrolyte

$$\bar{i} = -\kappa \nabla \phi$$

κ: conductivity; ϕ: electric potential

Electrolyte Conductivity

$$\kappa = F^2 \sum z_j^2 u_j c_j$$

z_j: charge on species; u_j: ionic mobility

$$u_j = \frac{D_j}{RT}$$ → Nernst Einstein equation

Electroneutrality

$$\sum z_j c_j = 0$$

↳ Excellent (and necessary) approximation everywhere

↳ Only on length scales (~nm) can charges sufficiently separate (@ electrode/electrolyte interface)